Lucas Systems Warehouse Pick Task Time Prediction

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1 Introduction

Warehouse efficiency is usually measured by how quickly and accurately workers can move products. One of the biggest challenges for managers is the unpredictability of human labor and behavior, which makes it hard to know exactly when a job will be finished. Lucas Systems tackles this problem with *Jennifer*, which is an AI voice-directed system that talks to workers through headsets. Jennifer tells people exactly which items to pick and where to go next, which helps keep things moving and reduces mistakes.

Even though Jennifer is capable of giving directions, it does not know exactly how long a full work queue will take. A work queue is a whole shift of work, filled with many pick tasks that are optimally chosen based on priority and location. Managers need to know the estimated time of completion of these queues so they can plan their staffing and schedules.

The goal of this project is to create a model that predicts how long these work queues will take by looking at data from three different warehouses. Two of these warehouses handle medical supplies, while the third focuses on hardware and power tools. Even though they house different products, they all use the same Jennifer software, which means the data they collect has the same structure. By looking at data like the distance between picks, the weight of the items, and how high up on the shelves the products are located, we can start to see patterns in how long each task takes.

One important note is that not all warehouse work is the same. Picking a small box for a conveyor belt is a much faster process than using a forklift to move a heavy pallet. Because of this, we are grouping the data by work codes. By treating different work codes as separate categories, we can build more accurate models for each specific type of task. Additionally, we have to deal with the fact that people are not robots, and don't follow the picking routine perfectly. Jennifer only tracks the moments when a worker actually finishes a task. It does not however see the time spent setting up equipment, taking a bathroom break, or stopping for lunch. This creates gaps in the data. To make our predictions as accurate as possible, we are cleaning the data to remove tasks that took an unusually long time, trying to account for the breaks. By focusing on the actual working time and retraining the model as new data comes in, we aim to give Lucas Systems a reliable way to predict a shift's work across any warehouse they serve.

2 Methods

2.1 Data

The data used in this exploration was provided to us by the Research & Development team at Lucas Systems. The data includes two customers and spans three warehouses for which we give each of the warehouses the following labels:

- OE: A warehouse located in Florida owned by a healthcare company managing medical equipment and supplies. The data spans `2025-09-08` to `2025-12-02`.
- OF: Another warehouse owned the same healthcare company as OE located in Maryland. The data spans `2025-09-06` to `2025-12-03`.
- RT: A warehouse located in Georgia owned by a hardware company focused on powertools and construction. This warehouse has two associated datasets, one named `RT_Activity` which spans `2024-05-28` to `2024-06-11`, and another called `RT_Activity2` going from `2023-10-03` to `2026-01-05`.

All activity in the warehouses is tracked by Jennifer which optimizes the sequencing of tasks to be completed and monitors progress real time. Workers wear headsets to communicate with Jennifer where they receive instructions on the next task to be carried out. With all warehouses using the same software for warehouse management the structure of the data is very consistent across warehouses, allowing for generalizable results in analyses and subsequent models.

The data pertaining to real time task completing by the workers in the warehouse is stored in files that we call `xx_Activity`, where `xx` is the particular warehouse. Each record in this data includes a Timestamp indicating when a particular task was completed by a worker in the warehouse. In order to quantify how long a specific task took to complete, we took the difference in Timestamps between two consecutive tasks completed by the same user. The data also includes rows indicating when workers started their assignments (indicated by `AssignmentOpen`). However, the start and end of assignments can warrant additional set up processes, which can bias task completion times. Therefore, `AssignmentOpen` rows and the first task in each assignment was removed from the analysis to combat this issue.

The dataset also includes other more static data. The locations of pick tasks are stored in a file that we called `xx_Locations`. This data maps `LocationID`s contained in the activity data to specific locations in the warehouse indicated by the Aisle, Bay, Level, and Slot numbers. The Aisle and Bay number can be thought as specifying a certain rack or shelf in the warehouse, Level indicates how high or low an item is, and Slot informs the worker specifically where in the row to look. The Aisle and Bay numbers were also used to calculate approximate distances between consecutive tasks using an associated distance matrix for the particular warehouse.

Finally, the dataset includes data on the particular products being managed by the warehouse which we named `xx_Products`. This file includes information on product weights and size, indicated by the volume (`Cube`) or a combination of `Length`, `Width`, and `Height`. Customers are free to choose one or the other, whether to simply provide the volume or the specific dimensions of each product. Customers are also free to choose whether or not to provide the units of the weight and dimensions, however, they are guaranteed to be consistent within the warehouse.

2.2 Data Preprocessing

The activity data records timestamps when workers complete tasks, so task duration was calculated as the time difference between consecutive tasks performed by the same user. Rows labeled `AssignmentOpen` were removed because they mark assignment starts and often include setup time that does not reflect actual picking effort. Extremely long time gaps were also excluded, as they likely correspond to breaks or interruptions that Jennifer does not observe.

The cleaned activity table was then joined with Product, Location, and distance matrices. Product attributes such as weight and volume were merged by `ProductID`, while location information (aisle, bay, level, slot) was joined using `LocationID`. Distances between consecutive tasks were calculated from warehouse-specific distance matrices. The resulting dataset integrates temporal, spatial, and product features for each task, enabling downstream modeling of completion time.

3 Methods

3.1 Models

Our goal is to predict the amount of time it takes for a particular work queue to be completed while maintaining a certain level of interpretability to inform customers on well-performing products

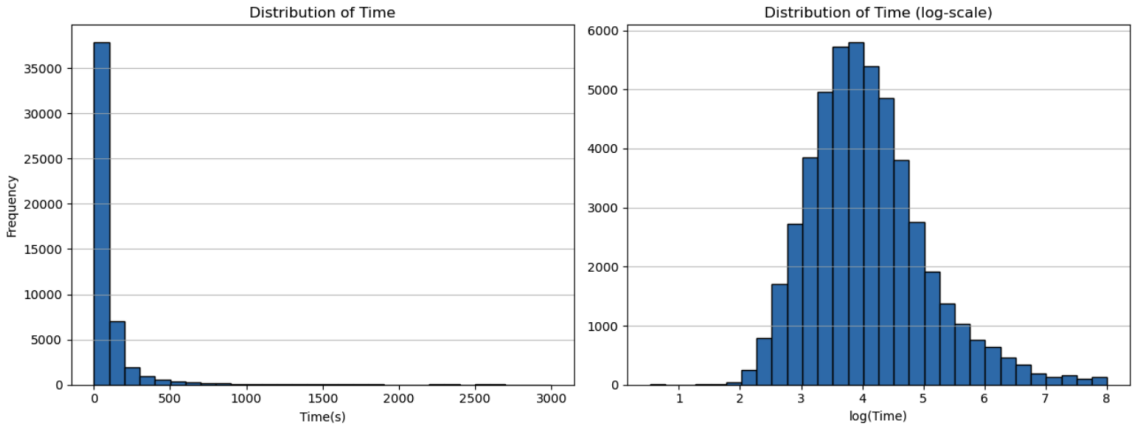

Figure 1: Raw time data between tasks is extremely right-skewed(left); after log-transformation, the variable is approximately normally distributed(right)

and factors leading to decreased work queue completion times. We largely consider three model frameworks: linear regression, decision trees, and neural networks.

Linear regression served as the baseline because its coefficients directly quantify how predictors influence time, making results easy to interpret. Decision trees were explored because they capture nonlinear effects and interactions automatically. They allow us to detect conditional patterns that linear models may miss.

4 Results

4.1 Exploratory Data Analysis

Since our primary task is predicting the completion time of warehouse pick tasks, the natural preliminary response to focus on would be the time taken for each individual task.

We first examined the distribution of task durations on both the original and logarithmic scales (see Figure 1). The raw distribution was highly right-skewed, with many short tasks around 1 second and a long tail of much longer tasks. After applying a log transformation, the distribution became more symmetric, suggesting that modeling log time is appropriate, especially for satisfying regression assumptions.

We also plotted log task time against log distance between consecutive picks, colored by work code with LOESS smoothing (see Figure 2). In general, there is a positive relationship: longer travel distances correspond to longer times in between tasks. However, trends differed across work codes, showing strong evidence of interactions between work types, distance, and time.

4.2 Model Performance

As a baseline, we fit a multiple linear regression model with time on the log-scale as response, and including features including distance (log-scale), product characteristics (weight, volume, quantity, all on log-scale), task type (WorkCode), spatial location (aisle and level, both as categorical variables), and temporal effects (time of day). The model achieves an R-squared measure of 0.338, indicating that it explains about 34 percent of the variation in task completion time. This model

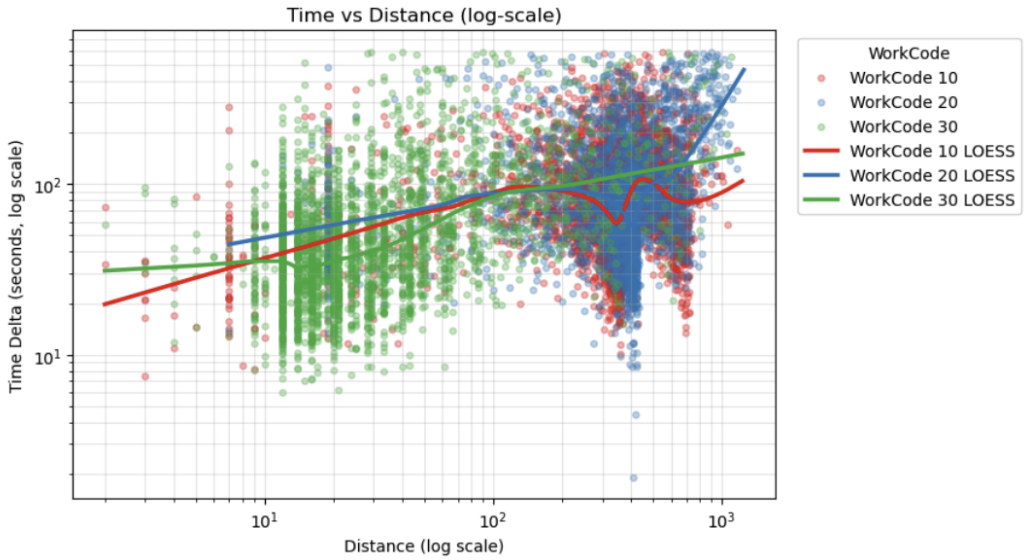

Figure 2: On the log-scale, there is a strong positive relationship between time and distance. The trends are similar across WorkCodes, though WorkCodes do differ by time and distance.

provides an interpretable benchmark against which more flexible models in future stages of this project.

5 Discussion

6 Appendix

Feature	dtype	n_missing	n_unique	min	max	mean
ActivityCode	object	0	2	NaN	NaN	NaN
UserID	int64	0	40	64.0	504.0	4.050262e+02
WorkCode	int64	0	3	10.0	30.0	2.642034e+01
AssignmentID	int64	0	42241	7717782.0	8042473.0	7.924902e+06
ProductID	float64	894	7691	1.0	57791.0	2.477778e+04
Quantity	float64	894	144	1.0	1143.0	8.453474e+00
Timestamp	object	0	95804	NaN	NaN	NaN
LocationID	float64	894	7669	1.0	8104198.0	8.932247e+05

Table 1: Activity Data Summary

Feature	dtype	n_missing	n_unique	min	max	mean
ProductID	int64	0	57671	1.0	57671.000	28836.000000
ProductCode	object	0	41981	NaN	NaN	NaN
UnitOfMeasure	object	0	55	NaN	NaN	NaN
Weight	float64	0	5572	0.0	7584.000	4.834848
Cube	float64	0	4009	0.0	421.296	0.623222

Table 2: Product Data Summary

Feature	dtype	n_missing	n_unique	min	max	mean
LocationID	int64	0	33519	1.0	8034868.0	795274.768400
Aisle	object	0	52	NaN	NaN	NaN
Bay	float64	1	86	1.0	99.0	23.107107
Level	float64	1	12	1.0	50.0	3.687153
Slot	float64	1	35	1.0	35.0	3.019243

Table 3: Location Data Summary